

05

Process-Modeling Foundations

Different representations answer different questions.

Four behavioral building blocks

Sequence

 $A \rightarrow B$

One follows another.

Choice

 $A \rightarrow (B | C)$

One branch is selected.

Parallel

 $A \rightarrow (B \parallel C)$

Both occur, order varies.

Loop

 $A \rightarrow B \rightarrow A$

Behavior repeats.

Key idea A useful model must distinguish choice from concurrency and permit repetition without permitting everything.

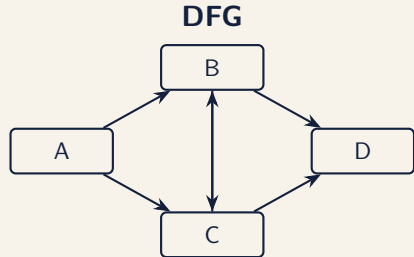
A DFG is not a complete process model

Observed traces

$\langle A, B, C, D \rangle$

$\langle A, C, B, D \rangle$

Evidence suggests B and C may be parallel.



The DFG also appears to allow repeated alternation between B and C . That behavior was never observed.

Model representations

Representation	Strength	Watch for
DFG	Fast, readable overview	No complete execution semantics
Process tree	Structured, sound by construction	Restricts model shape
Petri net	Precise concurrency and replay	Harder for non-specialists
BPMN	Familiar business notation	Semantics vary by usage
Transition system	State-based behavior	State explosion

Discuss Which representation would you use for stakeholder discussion? Which for alignment-based conformance checking?

Choose a modeling strategy

Purpose	Strategy	Representation	Emphasize
Explain flow	Structured discovery	Process tree / BPMN	Simplicity, fitness
Audit rules	Normative modeling	Petri net / Declare	Soundness, diagnostics
Explore complexity	Frequency-aware discovery	DFG / heuristics net	Coverage, readability
Predict state	State or prefix model	Transition / features	Generalization, utility
Communicate	Simplify and annotate	BPMN / DFG	Interpretability

Key idea There is no context-free “best model.” State the purpose before selecting the algorithm, representation, and quality criteria.

Abstraction before algorithm



Useful strategies

- ▶ Collapse technical events
- ▶ Separate process populations
- ▶ Decompose by lifecycle or subprocess

Practice

- ▶ Version every mapping
- ▶ Compare before and after profiles
- ▶ Retain a traceable raw-event link

Soundness: can every case finish properly?

For a workflow model, we usually want:

- ▶ One meaningful start and one meaningful completion
- ▶ Every modeled activity can occur
- ▶ No deadlocks before completion
- ▶ No tokens or work left behind after completion

Why it matters

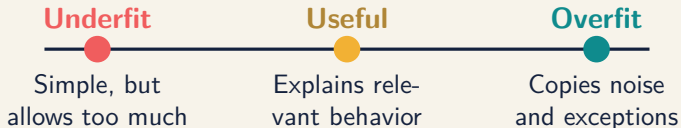
An attractive diagram can still encode impossible or incomplete behavior. Formal semantics let us test the model rather than trust its appearance.

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Process Discovery

Discovery algorithms compress event behavior into an explanatory model.

Discovery is a balancing act



Discovery depends on:

- ▶ The event log and its abstraction level
- ▶ The algorithm's behavioral assumptions
- ▶ Noise and frequency thresholds
- ▶ The intended use of the resulting model

Three discovery families

Alpha Miner

Uses ordering relations to infer causal, parallel, and choice behavior.

Best as a conceptual foundation.

Heuristics Miner

Uses frequency and dependency measures to tolerate noise.

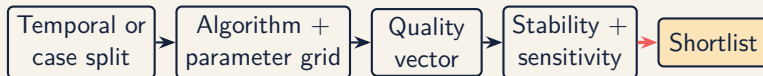
Useful for exploratory models.

Inductive Miner

Recursively finds sequence, choice, parallel, and loop cuts.

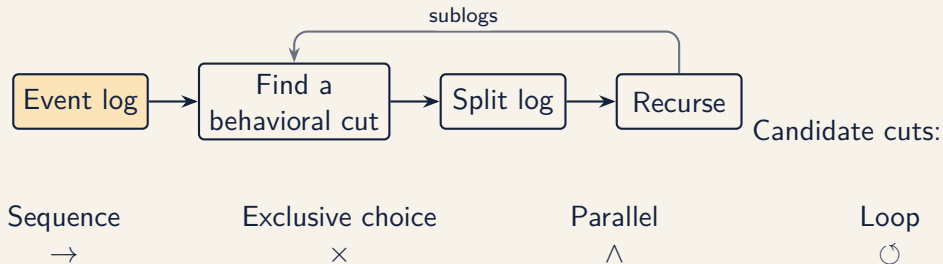
Produces sound, structured models.

A disciplined discovery experiment



1. Define constraints and metric directions before seeing results.
2. Keep discovery data separate from final evaluation data.
3. Retain all candidate parameters and artifacts, not only the winner.
4. Inspect unstable activities and edges, not only aggregate scores.

Inductive discovery intuition



Key idea The result is a process tree that can be translated into a Petri net or BPMN model.

Discover with PM4Py

```
import pm4py

# Structured discovery
tree = pm4py.discover_process_tree_inductive(
    log, noise_threshold=0.2
)
net, initial, final = pm4py.convert_to_petri_net(tree)

# Business-facing representation
bpmn = pm4py.convert_to_bpmn(tree)
pm4py.save_vis_bpmn(bpmn, "discovered_process.svg")
```

Experiment: vary the noise threshold. Record which behavior disappears, which remains, and whether the model becomes more useful.